

Imperial College London

BSc/MSci EXAMINATION Summer 2022

This paper is also taken for the relevant Examination for the Associateship

MRES - STATISTICAL METHODS FOR EXPERIMENTAL PHYSICS (MOCK EXAMINATION)

For MLBD MRes Students

June, 2022: 12:00 to 14:00

Answer all FOUR questions

Marks shown on this paper are indicative of those the Examiners anticipate assigning.

General Instructions

Complete the front cover of each of the 4 answer books provided.

If an electronic calculator is used, write its serial number at the top of the front cover of each answer book.

USE ONE ANSWER BOOK FOR EACH QUESTION.

Enter the number of each question attempted in the box on the front cover of its corresponding answer book.

Hand in 4 answer books even if they have not all been used.

You are reminded that Examiners attach great importance to legibility, accuracy and clarity of expression.

1. (i) A Bernoulli random variable X is a discrete random variable whose outcome is binary, often represented as a 0 or 1. The Bernoulli parameter p is the probability of ‘success’ or equivalently the probability that $X = 1$. We define it as $P(X = 1) = p$.

- (a) What is the expectation value of X when X is distributed as a Bernoulli random variable with parameter p ?

ANSWER: The expectation of X is given by,

$$E[X] = \sum_i X_i P(X_i) = 0 \times (1 - p) + 1 \times p = p$$

- (b) What is the variance of X when X is distributed as a Bernoulli random variable with parameter p ?

ANSWER:

The variance of X is given by,

$$V[X] = E[(X - E[X])^2] = E[(X - p)^2] = \sum_i (X_i - p)^2 P(X_i) = (-p)^2(1 - p) + (1 - p)^2 p = p(1 - p)$$

[5 marks]

- (ii) The binomial distribution describes the probability to observe k successful outcomes in n independent trials,

$$P(k; n, p) = \frac{n!}{k!(n - k)!} p^k (1 - p)^{n - k}$$

where p is the probability that a single outcome is successful.

- (a) Show that the expectation value of the binomial distribution is np

ANSWER: We can express the number of successful trials k as the sum over the outcomes X_i , $k = \sum_{i=1}^n X_i$. The expectation of k is given by,

$$E[k] = E\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n E[X_i] = np$$

- (b) Show that the variance of a binomial distribution is $np(1 - p)$. You can use the fact that the variance of the sum of independent variables is the same as the sum of the variances of the independent variables.

ANSWER: Similar to above,

$$V[k] = V\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n V[X_i] = np(1 - p)$$

[10 marks]

- (iii) We can estimate the binomial parameter p for a given observation of k and a fixed value n through both the method of moments and maximum likelihood estimator.

[This question continues on the next page ...]

- (a) Show that the method of moments estimator for p is given by $\hat{p} = \frac{k}{n}$.

ANSWER: The method of moments states that we should match moments of the distribution with sample moments. In this case, we can equate the following,

$$E[k] = \bar{k}$$

where $\bar{k}$ is the sample mean. Since there is a single value of k we must have that $\bar{k} = k$ and hence,

$$np = k \implies n\hat{p} = k \implies \hat{p} = \frac{k}{n}$$

- (b) What is the maximum likelihood estimator for p ?

ANSWER: The maximum likelihood estimator is the value of the parameter p for which,

$$\frac{dL(p)}{dp} = 0.$$

The likelihood $L(p)$ is given by,

$$L(p) = \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k}$$

and so our estimator $\hat{p}$ must satisfy,

$$0 = \frac{dL(p)}{dp} \Big|_{p=\hat{p}} = \frac{n!}{k!(n-k)!} \hat{p}^{k-1} (1-\hat{p})^{n-k} - \frac{n!}{k!(n-k)!} \hat{p}^k (n-k) (1-\hat{p})^{n-k-1}.$$

Rearranging we get,

$$\frac{k}{\hat{p}} = \frac{n-k}{1-\hat{p}} \implies \hat{p} = \frac{k}{n}$$

- (c) The “bias” of an estimator $\hat{\theta}$ is defined as the difference between the expectation value of the estimator and the value of the parameter being estimated θ ,

$$\text{bias}(\hat{\theta}) = E[\hat{\theta}] - \theta.$$

We say an estimator is “unbiased” if $\text{bias} = 0$. Show that the maximum likelihood estimator for the bernoulli parameter p is unbiased.

ANSWER:

$$E[\hat{p}] = E\left[\frac{k}{n}\right] = \frac{1}{n} E[k] = \frac{1}{n} np = p$$

hence the bias is $E[\hat{p}] - p = 0$ so the estimator is unbiased.

- (d) Name any property apart from unbiased that makes an estimator a “good” estimator. Describe what that property means.

ANSWER: Any of the following properties are considered as properties of good estimators; sufficient (minimal variance), consistent (converges in probability to the true value), invariant under change of variables and/or invariant under change of parameterisation.

[10 marks]

[Total 25 marks]

2. (i) Hypothesis testing is the frequentist process of rejecting (or accepting) a scientific hypothesis given some observation in experimental data. Formally two hypotheses are defined, H_0 := the null hypothesis, and H_1 := the alternate hypothesis.

- (a) State the definition of Type-I and Type-II errors in terms of the two hypotheses H_0 and H_1 .

ANSWER:

Answers can be given in the form of words or equations.

The Type-I error is the probability to reject the null hypothesis when the null hypothesis is true and the Type-II error is the probability to reject the alternate hypothesis (or accept the null hypothesis), when the alternate hypothesis is true.

$$\text{type - I error} = \alpha = P(X \in w | H_0)$$

$$\text{type - II error} = \beta = P(X \in W - w | H_1)$$

where $X \in W$ is some test statistic and w is the critical region.

- (b) A testing apparatus is designed to detect for a faulty pieces of lab equipment. The test will report 'T' if the equipment is faulty and 'N' if the equipment is fine. The designer claims that the probability that the test reads 'T' when equipment is faulty is 80%. You test out a piece of equipment in your lab with the apparatus and get a reading 'T'. Which of the following statements is true given the information you have.

- The probability that the equipment is faulty is 80%.
- If the equipment is not faulty, the probability that the test reports 'T' is 20%.
- The probability that the test would have read 'N' if the equipment was faulty is 20%

ANSWER: Only the third statement can be made given the information in the question.

[5 marks]

- (ii) An experiment is designed to detect decays of muon particles after being produced in an accelerator. A detector is setup to measure the decays time, t , after the the muon is produced $t_0 = 0$. The experiment runs until a maximum time T such that no decay times $t > T$ are recorded. The probability to observe a decay time t after is given by the exponential probability distribution,

$$f(t; \tau) = N \exp\left(-\frac{t}{\tau}\right)$$

where τ is the *lifetime* parameter.

- (a) Derive N as a function of τ

ANSWER: Using the property that the probability density is normalised to 1,

$$1 = \int_0^T N \exp\left(-\frac{t}{\tau}\right) dt = N \left[-\tau \exp\left(-\frac{t}{\tau}\right) \right]_{t=0}^{t=T} = \tau N \left(1 - \exp\left(-\frac{T}{\tau}\right) \right)$$

[This question continues on the next page ...]

hence,

$$N = \frac{1}{\tau \left(1 - \exp\left(-\frac{T}{\tau}\right)\right)}$$

- (b) The figure below (figure 1) shows the distribution a distribution in data (observed values of the decay time t) collected by the experiment and an exponential probability distribution.

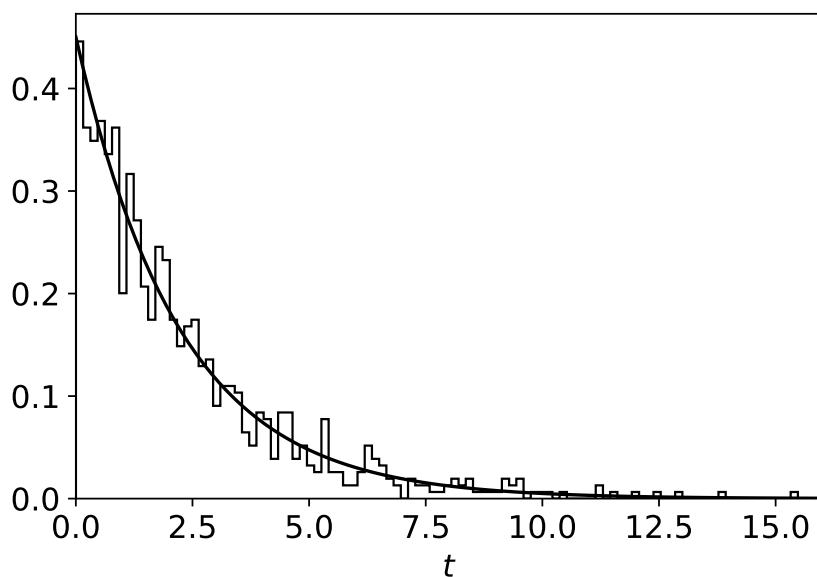


Figure 1

Describe how you would calculate the cumulative empirical distribution function $F_{emp}(t)$ for the data.

ANSWER: The empirical distribution function for the observed data is given by the following expression,

$$F_{emp}(t) = \frac{\text{number of observations smaller than } t}{\text{total number of observations}}.$$

An expression using the same bins as shown in the figure is also acceptable.

- (c) The Kolmogorov-Smirnov test-statistic is given by,

$$D = \max \left\{ |F_{emp}(t) - F(t)| \right\},$$

where $F(t)$ is the cumulative distribution of the exponential function in this case. Derive $F(t)$.

ANSWER: The answer is given by calculating the cumulative distribution function of the exponential probability density,

[This question continues on the next page ...]

$$F(t; \tau) = \frac{1}{\tau(1 - \exp(-\frac{T}{\tau}))} \int_0^t \exp\left(-\frac{t'}{\tau}\right) dt' = \frac{\tau(1 - \exp(-\frac{t}{\tau}))}{\tau(1 - \exp(-\frac{T}{\tau}))} = \frac{(1 - \exp(-\frac{t}{\tau}))}{(1 - \exp(-\frac{T}{\tau}))}$$

- (d) A test is performed to see if the exponential is a good description for the data shown. The figure below (figure 2) shows the distribution of D assuming the exponential shown is the true distribution ($f(D|H_0)$) and the observed value of the test statistic D_{obs} . The figure also shows a value D_{95} for which 95% of the distribution is to the left of the line. Describe how the information below could

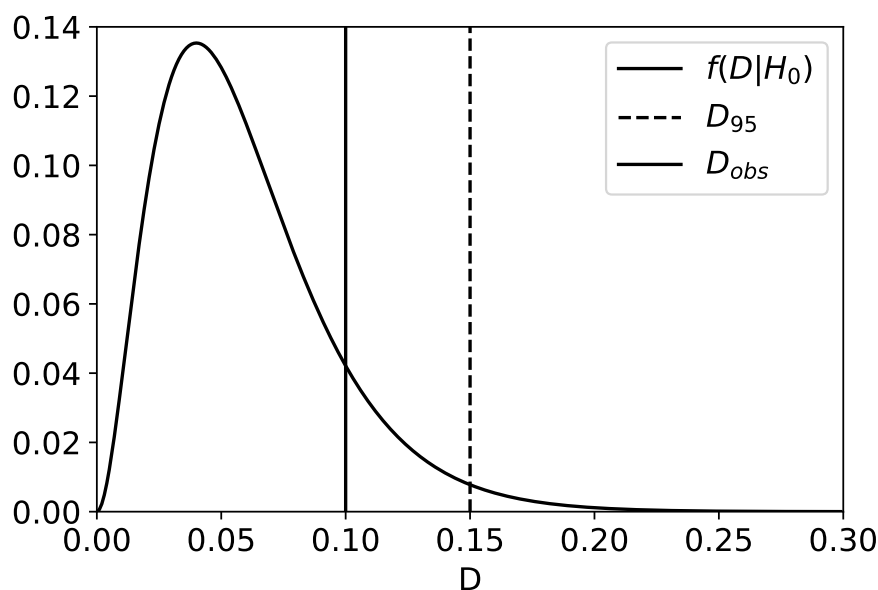


Figure 2

be used to test if the hypothesis that the exponential is a good description of the data is true or not.

ANSWER: The answer will depend on a threshold for the type-I error chosen by the experimenter. One answer could be that if the type-I error is chosen to be 0.05, since the observed value is below the threshold line, we would not reject H_0 .

[10 marks]

- (iii) Suppose now we would like to measure the parameter τ from the observed data using the same experiment as in part (ii). Assume for the rest of this question that the times measured are much smaller than the maximum time T or equally that $T \rightarrow +\infty$.

- (a) Write down the likelihood function for a set of n observations $\{t_1, t_2, \dots, t_n\}$ assuming the same exponential probability distribution from part (ii).

[This question continues on the next page ...]

ANSWER: Since $\rightarrow +\infty$, the probability density simplifies to,

$$f(t; \tau) = \frac{1}{\tau} \exp\left(-\frac{t}{\tau}\right).$$

The likelihood function is given by,

$$L(\tau) = \prod_{i=1}^n f(t_i; \tau) = \prod_{i=1}^n \frac{1}{\tau} \exp\left(-\frac{t_i}{\tau}\right)$$

- (b) Derive the maximum likelihood estimate of the parameter $\hat{\tau}$

ANSWER: The maximum likelihood estimate $\hat{\tau}$ is the value of τ for which the likelihood is maximised. This is the same as minimising the negative of the natural log of the likelihood.

$$q = -\ln L(\tau) = \sum_{i=1}^n \ln(\tau) + \sum_{i=1}^n \frac{t_i}{\tau} = n \ln(\tau) + \sum_{i=1}^n \frac{t_i}{\tau}$$

now, to find the minimum, we take the derivative of q with respect to τ and set to zero.

$$0 = \frac{dq}{d\tau}|_{\tau=\hat{\tau}} = \frac{n}{\hat{\tau}} - \frac{1}{\hat{\tau}^2} \sum_{i=1}^n t_i.$$

Multiplying through by $\frac{\hat{\tau}}{n}$, we have,

$$\hat{\tau} = \frac{1}{n} \sum_{i=1}^n t_i,$$

which is the maximum likelihood estimator for τ .

- (c) Explain how you would construct a Neyman interval at a 68% confidence level for the parameter τ . You should clearly state all of the steps required and you can write your answer in terms of pseudo-code if you wish.

ANSWER: The answer should contain the following set of steps:

1. Choose a test statistic t for the data. For example, this could be the based on log likelihood ratio $t = -2 \ln\left(\frac{L(\tau)}{L(\hat{\tau})}\right)$
2. Calculate the value of the test statistic for the data t_{obs} , which could depend on τ .
3. For a range of τ values derive the distribution $f(t; \tau)$ of the test-statistic assuming τ is the true value. Choose a procedure to construct 68% of the distribution in t for each value of τ . Any well motivated procedure is ok , for example the likelihood ratio ordering principle is ok, or if the test-statistic is the one based on the log likelihood ratio, largest values could be included first.

[This question continues on the next page ...]

4. For each value of τ determine whether the observed value t_{obs} lies inside the 68% region or not. This could be based on calculating a p-value $p = \int_{t_{obs}}^{\infty} f(t; \tau) dt$ and choosing values of τ for which $p > 0.68$.
5. Join the values chosen to form an interval in τ .

[10 marks]

[Total 25 marks]

3. (i) (a) State Bayes theorem, naming all the terms when it is used for parameter inference of model parameters θ with some data d .

ANSWER:

$$p(\theta|d) = \frac{p(d|\theta)p(\theta)}{p(d)}$$

$p(\theta|d)$ is posterior, $p(d|\theta)$ is likelihood, $p(\theta)$ is prior. $p(d)$ is evidence.

- (b) In what circumstances might you consider using a 'Jeffreys' prior $\pi(\theta) \propto 1/\theta$?

ANSWER: If θ is a scale parameter, i.e. the likelihood is expected to be invariant to multiplicative changes in θ (or a translation in $\ln(\theta)$), a prior uniform in $\ln(\theta)$ is natural, or $\propto 1/\theta$.

[6 marks]

- (ii) An experiment is conducted to measure the number of photons coming from a reactor. The counts are Poisson distributed

$$p(n|\lambda) = \frac{\lambda^n e^{-\lambda}}{n!}$$

where λ is the expected number during the experiment.

- (a) Formulate this as a Bayesian inference problem to infer the rate λ given a measurement n . With a Jeffreys prior for λ find an expression for the posterior.

ANSWER:

$$p(\lambda|n) \propto p(n|\lambda)\pi(\lambda) \propto \frac{\lambda^n e^{-\lambda}}{n!} \frac{1}{\lambda}.$$

- (b) What is the most probable (MAP) value of λ ?

ANSWER: Differentiate $\ln p(\lambda|n)$ (for convenience):

$$0 = \frac{d}{d\lambda}(n-1)\ln\lambda - \lambda \Rightarrow 0 = \frac{(n-1)}{\lambda} - 1 \Rightarrow \lambda_{\text{MAP}} = n-1.$$

- (c) Write down an expression for $\bar{\lambda}$, the posterior mean of λ in terms of $p(\lambda|n)$, and show that

$$\bar{\lambda} = n.$$

You may use the result $\int_0^\infty \lambda^n e^{-\lambda} d\lambda = n!$

ANSWER: The posterior mean is

$$\bar{\lambda} = \int p(\lambda|n) \lambda d\lambda$$

Since we have to normalise the posterior, we write in terms of the unnormalised posterior we found before:

$$\bar{\lambda} = \frac{\int p(\lambda|n) \lambda d\lambda}{\int p(\lambda|n) d\lambda}$$

Using the given result, this is

$$\bar{\lambda} = \frac{\int_0^\infty \lambda^n e^{-\lambda} d\lambda}{\int_0^\infty \lambda^{n-1} e^{-\lambda} d\lambda} = \frac{n!}{(n-1)!} = n.$$

[This question continues on the next page ...]

[8 marks]

- (iii) Find the variance of the posterior.

ANSWER: The posterior expectation of λ^2 is

$$\langle \lambda^2 \rangle = \frac{\int_0^\infty \lambda^2 \lambda^{n-1} e^{-\lambda} d\lambda}{\int_0^\infty \lambda^{n-1} e^{-\lambda} d\lambda} = \frac{(n+1)!}{(n-1)!} = n(n+1).$$

Hence the variance is $\langle \lambda^2 \rangle - \bar{\lambda}^2 = n(n+1) - n^2 = n$.

[4 marks]

- (iv) The Fisher matrix for parameters θ_α is

$$F_{\alpha\beta} \equiv \left\langle -\frac{\partial^2 \ln L}{\partial \theta_\alpha \partial \theta_\beta} \right\rangle,$$

where L is the likelihood. In this case the matrix is a single number, whose inverse gives the (minimum) expected value of the variance.

Show that the prediction for the variance from the Fisher analysis is $\sigma_\lambda^2 = n - 1$, and give two reasons for why it differs from the variance you calculated from the posterior.

ANSWER: Take the log of the likelihood:

$$\ln L = n \ln \lambda - \lambda + \text{cst.}$$

Hence

$$\frac{\partial \ln L}{\partial \lambda} = \frac{n}{\lambda} - 1 \Rightarrow \frac{\partial^2 \ln L}{\partial \lambda^2} = -\frac{n}{\lambda^2}.$$

Hence the Fisher matrix (scalar) is

$$F = \left\langle -\frac{\partial^2 \ln L}{\partial \lambda^2} \right\rangle = n \langle \lambda^{-2} \rangle = \frac{\int_0^\infty \lambda^2 \lambda^{n-2} e^{-\lambda} d\lambda}{\int_0^\infty \lambda^n e^{-\lambda} d\lambda} = n \frac{(n-2)!}{n!} = \frac{1}{n-1}$$

using the same method as before. Hence the Fisher prediction for the variance is $n - 1$ which differs from the previous result because it is based on the likelihood, not the posterior, and because it is computed from the second derivative at the peak, not on an exact integration for the variance.

[7 marks]

[Total 25 marks]

4. A measurement is made of a galaxy's recession speed v , to determine whether the Universe is expanding or not. Two models are considered:

- ssm: The Steady-State model: $v = 0$.
- bbm: The Big Bang model: $v = Hr$,

where r is the distance to the galaxy, which we assume that we know ($r = 1$ Mpc), and H is the Hubble constant, a parameter of the model. Throughout this question, use units of km s^{-1} for velocities, Mpc for distances, and $\text{km s}^{-1}\text{Mpc}^{-1}$ for H . Unlike normal practice, it is not necessary to quote the units for the solutions.

The recession speed of the galaxy is measured, giving $v = 90 \text{ km s}^{-1}$. It is known that the measurement error distribution is gaussian, with zero mean and standard deviation $\sigma = 30 \text{ km s}^{-1}$.

- (i) (a) The models are 'nested'. Explain what this means and why it applies in this case.

ANSWER: One model is a special case of the other. [2]

- (b) Assuming a uniform prior, what is the posterior of H , in the bbm? There is no need to normalise it at this stage,

ANSWER:

$$p(H|v) \propto p(v|H) \propto \exp\left[-\frac{(v - Hr)^2}{2\sigma^2}\right]$$

(with $r = 1$, $v = 90$, $\sigma = 30$ in the units given). [3]

[5 marks]

- (ii) (a) What is the most probable value of H in the bbm?

ANSWER: $H = v$. [2]

- (b) Calculate the posterior uncertainty on H .

ANSWER: The posterior is gaussian in H for fixed data v , so, by inspection, the variance is σ^2 . [3]

[5 marks]

- (iii) Now we consider model selection, to see whether the data prefer the ssm or the bbm. For this we need to pay attention to the normalisation of the probability distributions.

- (a) Assume that the uniform prior on H extends from -200 to $+200 \text{ km s}^{-1}\text{Mpc}^{-1}$ and is zero outside this range. What is the (normalised) prior?

ANSWER: $\pi(H) = 1/400$ for $-200 < H < 200$, 0 otherwise. [2]

- (b) Write down the product of the likelihood and the prior, now properly normalised.

ANSWER:

$$p(H|v) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(v - Hr)^2}{2\sigma^2}\right] \times \frac{1}{400}$$

for $-200 < H < 200$, and zero otherwise. [2]

[This question continues on the next

[4 marks]

- (iv) (a) Hence find an integral expression for the Bayesian Evidence for the bbm, and integrate it to show that

$$Z_{\text{bbm}}(v) \simeq \frac{1}{400},$$

where you should explain why it is approximate, and why the approximation will be accurate in this case.

ANSWER: The evidence is the product of the likelihood and the prior, integrated over the parameter H :

$$\begin{aligned} Z(v) &= \int_{-200}^{200} \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(v-Hr)^2}{2\sigma^2}\right] \times \frac{1}{400} dH \\ &\simeq \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(v-Hr)^2}{2\sigma^2}\right] \times \frac{1}{400} dH. \end{aligned} \quad (1)$$

For this to be valid, we assume that the integrand is very small outside the prior range, which is justified since the peak is at $H = 90$, and the standard deviation is $\sigma = 30$, which is much smaller than 110, which is the distance from the peak to the nearest prior boundary. [5]

- (b) Argue that in the ssm, the Bayesian Evidence is simply the likelihood.

ANSWER: The evidence is just $p(v)$, which is just $p(v|H = 0)$ in the ssm. This is just the likelihood.[2]

- (c) Hence calculate the Bayes factor $B_{\text{bs}} \equiv Z_{\text{bbm}}/Z_{\text{ssm}}$ and say which model is favoured by the data.

ANSWER: The Bayes factor is

$$B_{\text{sb}} = \frac{Z_{\text{ssm}}}{Z_{\text{bbm}}} = \frac{(2\pi\sigma^2)^{-1/2} \exp[-v^2/(2\sigma^2)]}{1/400} = \frac{400}{\sqrt{2\pi}30 \exp(9/2)} = 0.059.$$

Hence the Big Bang model is favoured with relative odds of about 16. [4]

[11 marks]

[Total 25 marks]